

RESONANCE: Frequency-Aware GRACE-fully Corrected Pixel Features for Robust AIGC Image Detection Under Real-World Transformations

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Abstract

Detectors for AI-generated images are often trained and evaluated on clean inputs, yet in real-world deployment, images are often transformed — re-encoded, blurred, rescaled, or resized, before they are inspected. This work investigates whether performance collapse of detectors fit on pristine images can be repaired by a lightweight, label-free module, without requiring full retraining of the underlying detector. This work introduces GRACE, a gated residual adapter that maps the internal feature representation of a frozen detector toward the representation that would have been produced from a clean image. The adaptation is conditioned on a predicted degradation severity and initialized as an identity mapping, ensuring that any subsequent change in the detector’s output can be attributed to what the module learns during training. The approach is further complemented by a frequency-domain branch that re-examines the same image window using a per-block DCT basis, preserving high-frequency generation traces that may survive spatial resizing even when other discriminative information is degraded. This complementary evidence is integrated through band-gated cross-attention. On a held-out benchmark of COCO reals and DALL-E 3 fakes, this work evaluates AUC of images degraded across an eleven-transformation degradation grid, AUC of clean images and additionally computes a retention metric which is defined as degraded AUC normalized by the detector’s own clean-image AUC.

1 Introduction

Generative image models now produce images that are difficult to distinguish from photographs, and detectors trained on their artifacts have become a standard safeguard. Most such detectors, however, are fit and benchmarked on clean inputs. The gap between that setting and deployment is ordinary handling: an image is re-encoded by a social media or messaging platform, blurred by a camera, or resized into a thumbnail before a detector is able to see it. A detector whose evidence lives in local high-frequency structure collapses under exactly these operations.

This work asks whether the collapse of useful artifacts under typical image transformations is *repairable*. Specifically, a frozen detector trained on clean images is used as the starting point, and a lightweight module is inserted between its backbone and frozen classification head to recover lost performance without requiring image-level labels. Beyond repairing the detector’s internal representation, this work investigates whether information discarded by the backbone can be recovered by re-reading the image in a complementary basis—the frequency domain.

The contributions of this work are as follows:

1. **GRACE** — a Gated Residual Adapter for Clean-feature Estimation that restores a degraded feature vector toward its clean counterpart through a frozen head, trained label-free on the detector’s own clean features and the degradation sampler’s own metadata (Sec. 2). It is identity at initialization, per-channel gated, and small — roughly 0.3M parameters against a 21M frozen trunk.
2. A **frequency enricher** that re-reads the image at native pixel scale via a per-block DCT, and fuses frequency-domain evidence with the restored feature through band-gated cross-attention (Sec. 2.4).

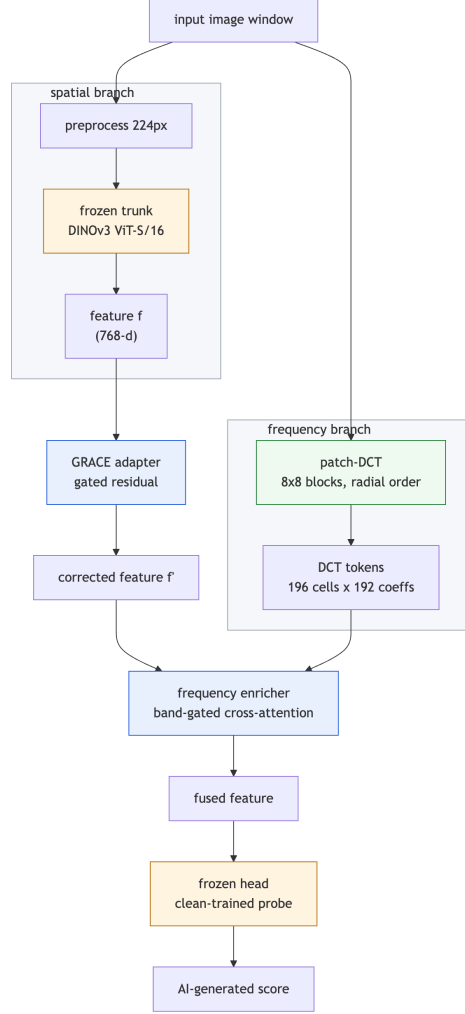


Figure 1: High-level pipeline. The same image window is read twice: a spatial branch (frozen trunk $\rightarrow$ feature f) and a frequency branch (patch-DCT $\rightarrow$ DCT tokens). The severity head predicts drift s ; the GRACE adapter restores f to f' ; the frequency enricher fuses DCT evidence into f' ; the frozen head produces the score.

2 Method

2.1 Overview

The pipeline wraps a frozen detector head $\circ$ trunk trained on clean images. The trunk is a frozen DINOv3 ViT-S/16 [1] embedding a 224px tensor into a 384-dimensional feature f ; the head is a frozen MLP probe mapping f to a single logit. Two corrective components are inserted at the seam f : a **severity head** that predicts how far the image has drifted from clean, a **GRACE adapter** that restores f , and a **frequency enricher** that adds frequency-domain evidence the resize destroyed. The key idea is that the image is read twice: once as a normalized 224px tensor for the spatial branch, and once at native pixel resolution for the frequency branch. Figure 1 shows the full pipeline.

2.2 Degradation model

Robustness here is measured against real-world handling of images. This work uses a pre-defined grid of eleven real-world image transforms (JPEG re-encode, Gaussian blur, resize, Gaussian noise, brightness /

contrast / saturation $\pm 20\%$, center crop), each with an explicit parameter list and a real-world analog. The six photometric entries are colour jitter unbundled into individual $\pm 20\%$ steps so each is its own attributable condition. Every transform is size-preserving and deterministic given its parameter (only Gaussian noise draws randomness), so the grid composes in any order.

Transforms for training data are applied at four severity levels:

1. **L0** clean (reference);
2. **L1** a single transform at one parameter (a 19-condition controlled sweep, one factor at a time);
3. **L2** a pair of distinct transforms; and
4. **L3** three to five distinct transforms.

The four-level augmentation pipeline is an intentional design choice, inspired by the degradation pipeline introduced by the Ant International team in the NTIRE 2026 Robust AI-Generated Image Detection challenge [3]. Rather than optimizing for a single transformation in isolation, the pipeline exposes the model to progressively diverse combinations of degradations, encouraging generalization beyond the individual transformations used during evaluation.

Each draw is keyed on (image index, level, replicate) so the degraded set is identical across runs, machines, and detectors. Critically, the severity of each draw is computable for free: a transform’s parameter is ranked within its own (mild→severe) grid and combined with composition depth into a scalar $s \in [0, 1]$. This is the supervision signal for the severity head, and it carries no image label.

2.3 GRACE adapter

The stage-1 component is a **gated residual adapter** that corrects a degraded feature back toward what a clean image would produce:

$$f' = f + g \odot \text{MLP}(\text{LN}(f)), \quad g = \sigma(\text{gate_logit}), \quad (1)$$

where $\odot$ is per-channel multiplication and the MLP is a stack of bottleneck-width residual blocks (bottleneck = 128). The per-channel gate g decides how much of each proposed correction to apply, and is modulated by the predicted severity s through FiLM [4], so a mildly degraded image gets a light touch and a heavily degraded one gets more. Two design requirements contribute to the method’s validity:

1. **Identity at initialization.** The last projection of every block is zero-initialized, so the adapter returns its input bit-for-bit at step 0 whatever the gate or severity. Any clean-AUC change is therefore attributable to what the adapter learned, not to its wiring.
2. **Gated residual.** The gate starts at $\sigma(-4) \approx 0.018$, so the adapter is a near-no-op and must learn to open; this prevents over-correction of clean inputs.

Figure 2 shows one residual block of the adapter.

2.4 Frequency enricher

The adapter can only rearrange what the trunk kept; a ViT at 224px has already resampled the window, and generation traces are a local high-frequency phenomenon at native pixel scale. The frequency branch therefore reads the pixels the trunk threw away, at native pixel scale (the same window in the window-aligned variant; the full native image in the native-resolution variant). Its tokens come from a **patch-DCT**: per- 8×8 -block DCT-II (8 being the JPEG block size), coefficient magnitudes pooled onto a fixed 14×14 cell grid and log-compressed — one token per DINOv3 patch cell, 192 radially-ordered coefficients per cell.

Two **band experts** (low- and high-frequency) run multi-head cross-attention with the corrected feature f' as query and the DCT cells as keys/values, each expert masked to its own contiguous slice of the radial coefficient axis. Different transforms interact differently with the frequency domain — blur and downscale destroy high-frequency energy, Gaussian noise adds it, and JPEG moves energy onto block-aligned coefficients,

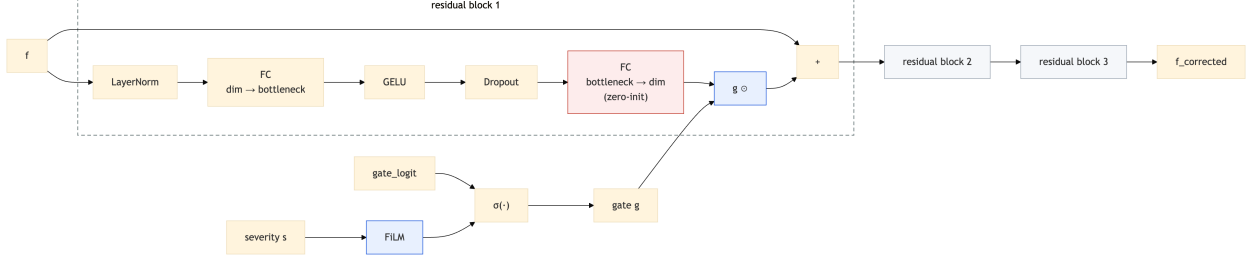


Figure 2: The GRACE adapter. Three residual blocks run in sequence; the first is shown in detail inside the dashed box. Within a block a LayerNorm→FC→GELU→FC path proposes a correction, the zero-initialized last projection guaranteeing identity at initialization. A per-channel gate $g = \sigma(\cdot)$, FiLM-modulated by severity s , scales the correction added back to f .

so a single expert with one scalar gate cannot open for one kind of damage and shut for another. The experts’ outputs are gated and added as a residual:

$$f_{\text{fused}} = f' + \sum_b g_b \cdot \text{BandExpert}_b(\text{tokens}, f'), \quad (2)$$

with each g_b a per-channel gate also FiLM-conditioned on s . Like the adapter, every expert’s output projection is zero-initialized, so $f_{\text{fused}} = f'$ exactly at initialization. Figure 3 details one band expert and the fusion.

2.5 Training objective

The adapter is trained *label-free*: it pulls degraded features onto the detector’s own clean features through the frozen head. The alignment loss is

$$\mathcal{L}_{\text{align}} = w_{\text{cos}} \cdot (1 - \cos(f', f_{\text{clean}})) + w_{\text{err}} \cdot \mathcal{E}(f' - f_{\text{clean}}), \quad (3)$$

where cosine and squared error are both kept because the head is sensitive to feature magnitude. By default the error term is **Jacobian-weighted**:

$$\mathcal{E} = (1 - \varepsilon) \mathbb{E}_B[(\hat{j} \cdot e)^2] + \varepsilon \mathbb{E}[e^2], \quad \hat{j} = \nabla_f h(f)|_{f_{\text{clean}}}, \quad (4)$$

which spends adapter capacity on the feature directions that move the head’s decision, with an isotropic floor $\varepsilon = 0.01$ (the reference arm uses $w_{\text{err}} = 4$, $w_{\text{cos}} = 0.25$, i.e. a 16:1 ratio). A head-KL term (the exact finite-difference counterpart) and a severity regression term are added with small weights.

The frequency enricher is trained in a separate second stage against the *frozen* stage-1 adapter (severity head included), and it is the one place the pipeline uses image labels. Its objective is

$$\mathcal{L}_{\text{enrich}} = \text{BCE}(h(f_{\text{fused}}) + \beta a(\Delta), y) + \lambda_{\text{aux}} \text{BCE}(a(\Delta), y) + \lambda_{\text{orth}} \text{orth}(\Delta, \cdot), \quad (5)$$

where f_{fused} is the gated enrichment, $f_{\text{fused}} = f' + \sum_b g_b \text{BandExpert}_b(\text{tokens}, f')$, and Δ is the DCT branch’s *un-gated* read, $\Delta = \sum_b \text{BandExpert}_b(\text{tokens}, f')$. The fused logit adds a learned scalar β times an auxiliary head $a(\cdot)$ (a small MLP); β starts at zero, so the step-0 prediction is exactly the GRACE arm’s. The two regularizers both read Δ on purpose, since a loss on the gated sum could be satisfied by closing a gate: the aux head forces the frequency read to be label-predictive on its own, and the orthogonality term (squared cosine) pushes Δ into directions the spatial feature does not already span (against the frozen head’s decision direction by default), so the enricher must carry forensic content the spatial seam lacks. The earlier anchor term $\|f_{\text{fused}} - f'\|$ survives only as an ablation and is off by default, and the reference configuration fine-tunes a copy of the detector head (with a differential warm-up schedule) alongside the enricher. As in stage 1, the enricher is scored before the first optimizer step, when every expert’s output projection is still zero, so the fused AUC must equal the GRACE arm’s exactly — a wiring check that is measured, not asserted.

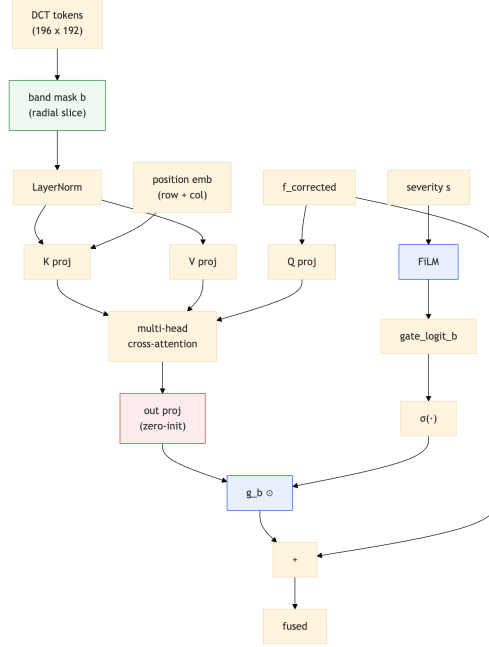


Figure 3: The frequency enricher: band-gated cross-attention fusion. DCT tokens are masked to a band, normalized, and projected to keys/values (position added to keys only); the corrected feature f' is the query. A zero-initialized output projection and a severity-modulated gate g_b gate each band’s contribution into the residual sum.

3 Experiments

3.1 Data

A weighted subset of the WildFake corpus is sampled (ModelScope, Apache 2.0): 49,999 base training images (14,191 real and 35,808 fake) and a balanced 10,000-image validation split. Because the corpus is dominated by older Stable Diffusion material, it is extended with supplementary single-class sources covering recent state-of-the-art DiT generators: 6,000 FLUX images (train) and 2,000 (validation) from FLUX-Reason-6M (HuggingFace, Apache 2.0), and 200 Seedream 4.5 images from seedream-4.5-generated-2k (HuggingFace, MIT). To hold the class balance, 2,457 (train) and 2,000 (validation) additional real photographs are sampled from LAION-5B and ImageNet, disjoint from the reals already in the manifest. The combined training corpus is therefore 58,656 images (16,648 real / 42,008 generated) with a balanced 14,000-image validation split (7,000 / 7,000). The fake half remains heavily weighted toward recent diffusion and DiT models over GANs (Stable Diffusion, Midjourney, FLUX, and Seedream 4.5 dominate, with a minority of GANs). The primary benchmark is a held-out 13,841-image set of COCO val2017 reals (4,998) and DALL-E 3 fakes (8,843); training and test are disjoint by construction (DALLE and COCO are excluded from training). Degradation is applied at native resolution, after which a multi-scale crop (128–256px, drawn deterministically per image) is taken before the 224px squash by the spatial branch.

3.2 Protocol and metric

All detectors are scored on a single, shared *condition lattice* of 26 conditions: clean, the 19 L1 grid points, and L2/L3 at three independent replicates each. The lattice is seeded on the image index, so every detector sees byte-identical degraded images and a difference between two detectors is never a difference in the draw. Clean is scored first, and every later condition scores the same images in the same order, so clean and degraded scores are paired per image. Clean scores additionally fix the operating point: a threshold is selected on clean data (max F_1) and applied unchanged to every degraded condition, so the harness exposes the calibration drift that AUC hides and reports a per-condition false-positive / false-negative breakdown at that

fixed threshold. The composed (L2/L3) levels are Monte-Carlo estimates, so their AUC is reported with a bootstrap confidence interval that resamples images rather than rows, and each image’s recipe is logged so the composed levels can be analysed per transform set.

All headline runs score a single deterministic arm, **crop200**: a 200×200 window at native pixel scale, the largest window every benchmark image can supply from its own pixels (the COCO reals ship at exactly 200×200 , while the fakes are native 1024px). Within the arm every image has identical dimensions, so the dimension shortcut — which scores AUC 1.0 on the raw benchmark — is 0.5 by construction rather than by normalization. This is the informative arm for the frequency branch.

The headline metric is *retention*, defined *chance-corrected* so that a detector collapsing to chance scores 0 rather than 0.5:

$$R = \frac{\text{AUC}_{\text{degraded}} - 0.5}{\text{AUC}_{\text{clean}} - 0.5}, \quad (6)$$

and reported comparisons are made against a shared denominator — the *baseline’s* clean AUC — rather than each arm’s own. A restored feature is bounded by the clean-feature score, but the frequency branch adds information that is not present in the clean image, changing both the numerator and the denominator; normalizing by an arm’s own clean AUC would hide exactly the effect being claimed.

3.3 Baselines and ablations

Baselines. The unadapted detector’s retention curve is the denominator every arm is compared against. A null-adaptor gate (`head(adapter(trunk(x)))` with `checkpoint: null`) must reproduce the baseline to the last decimal, guaranteeing the trunk/head split is sound. The frequency branch has an analogous gate: a freshly-initialized enricher must reproduce the +GRACE arm exactly, proving the fused wiring is measured rather than asserted. Two floors sit below the detector’s number and are reported alongside it: a metadata-only logistic regression on width, height, byte size, and bits-per-pixel — the “container floor,” which a real result must clear to be evidence about forensics rather than file properties — and a spectral-rolloff analysis on the crop200 arm, since the 200×200 reals are downsamples whose spectra roll off at the old Nyquist while the fake windows are untouched native pixels.

Stage-1 ablations. Each is the reference arm with exactly one key changed: plain MSE (Jacobian weighting off), doubled bottleneck (256 vs 128), reduced residual blocks (2 vs 3), gate initialization (-3 vs -4) and gate weight-decay exemption, and the $w_{\text{err}}/w_{\text{cos}}$ ratio swept over $\{0.25, 1, 4, 16\}$ (the reference sits at 16). Because the headline numbers come from a single seed, repeated runs are additionally aggregated into mean and confidence intervals via `seed_stats`, so a reported gap can be read against the seed-noise floor.

4 Results

All arms are scored on the held-out benchmark (COCO val2017 reals + DALL-E 3 fakes, 13,841 images) under the crop200 arm, with the chance-corrected retention of Eq. (6). Clean is scored first and fixes the operating threshold ($\max F_1$), which is then applied unchanged to every degraded condition. The reported values are the harness’s, and both identity gates of Sec. 3.3 held: the null adapter reproduced the baseline and a freshly-initialized enricher reproduced the +GRACE arm exactly (step-0 $\text{AUC}_{\text{corrected}} = \text{AUC}_{\text{fused}}$ throughout stage 2).

4.1 Headline results

Table 1 summarizes the three arms across severity levels. Table 2 isolates the clean-versus-transformed comparison the problem statement asks for, and Table 3 breaks the L1 sweep down by transformation, reporting each transform’s worst parameter point.

4.2 Results analysis

Representative false positives. The dominant error at the fixed threshold is over-triggering on degraded *reals*, concentrated on detail-destroying transforms. On clean data the baseline already calls 18% of reals

Detector	Clean AUC	Retention			Env.
		L1	L2	L3	
Baseline	0.9569	0.9083	0.8703	0.7507	3
+GRACE	0.9604	0.9588	0.9390	0.8697	3
+GRACE-freq	0.9719	0.9611	0.9452	0.8859	3

Table 1: Clean AUC and chance-corrected retention per severity level on crop200 (wildfake-coco-dalle3). Env. = operating envelope: the deepest composition level still at or above retention 0.5. The bootstrap CIs on the composed levels are non-overlapping across arms at every level (e.g. L1: [0.911, 0.918], [0.938, 0.945], [0.950, 0.956]).

Detector	Clean			Transformed (L1)		
	AUC	FPR	FNR	AUC	FPR	FNR
Baseline	0.9569	0.181	0.056	0.9150	0.237	0.098
+GRACE	0.9604	0.165	0.061	0.9414	0.212	0.076
+GRACE-freq	0.9719	0.138	0.046	0.9536	0.208	0.056

Table 2: Clean versus transformed performance on wildfake-coco-dalle3. “Transformed” pools the 19 single-transform L1 conditions; FPR and FNR are measured at the fixed clean-chosen threshold (max F_1). The adaptation narrows the clean-to-transformed gap on every row, and the frequency branch does so while also *raising* clean AUC and lowering clean FPR.

generated at the max- F_1 point; after a $0.25\times$ downscale it calls 82%, and after Gaussian blur ($\sigma=2.0$) 48%. These are the representative case: a real photograph whose texture is smoothed away reads like a diffusion output, because both are low-detail and spectrally rolled off. Mild photometric brightening or contrast increases nudge the same error up a few points. The adaptation roughly halves the resize false-positive rate ($0.82 \rightarrow 0.37$). The most likely mechanism is that these transforms erase the local high-frequency texture the detector keys on, moving a real photograph into the same smooth, spectrally-rolled-off region of feature space as a generated image; the correction can remap the feature back but cannot resurrect detail that no longer exists in the pixels, so a residual false-positive bias persists.

Representative false negatives. False negatives concentrate where the fake’s generation trace is *hidden*. Gaussian noise ($\sigma=0.1$) raises the baseline’s false-negative rate to 38% — the additive energy masks the high-frequency signature on the fake side — and heavy JPEG (quality 30) to 12%. The corrected arms cut the noise false-negative rate to 7–9%. The likely mechanism is that diffusion images are distinguished largely by being *too smooth* — abnormally little high-frequency energy — and additive noise supplies exactly that missing energy, so a corrupted fake’s spectrum becomes statistically similar to a real camera’s sensor noise; the DCT branch reads the same inflated coefficient magnitudes and cannot separate “generated plus noise” from “camera plus sensor noise” in the frequency domain. The asymmetry is informative: noise hides fakes (false negatives), while resize and blur manufacture fake-like reals (false positives); both directions act on the same high-frequency evidence.

Trade-offs surfaced by the errors. Three patterns are worth naming because they are not predictions of the design. First, the frequency branch is not uniformly better (Table 3): it slightly regresses on the worst JPEG and noise conditions relative to the spatial-only adapter ($0.898 \rightarrow 0.878$ and $0.855 \rightarrow 0.849$), because block-aligned JPEG energy and broadband noise occupy the same bands its experts read, while winning decisively where the spatial read is destroyed (resize $0.838 \rightarrow 0.876$, blur $0.896 \rightarrow 0.922$). Second, the repair largely removes the *interaction effect*: the unadapted baseline’s L3 retention (0.751) sits far below its independence prediction (-0.058 interaction gap), whereas the corrected arms are close to additive (gap $+0.004$ at L2 and -0.001 at L3 for the fused detector) — single-transform sweeps alone understate the baseline’s failure, and the repaired detector behaves as if degradations compose independently. Third, a clean-to-transformed gap persists even for +GRACE-freq ($0.972 \rightarrow 0.954$ at L1, 0.918 at L3), and the fixed-threshold FPR stays above its clean value (20.8% vs 13.8%): the correction narrows but does not remove

Transform (worst parameter)	Baseline	+GRACE	+GRACE-freq
resize (scale 0.25)	0.5024	0.8381	0.8756
gaussian blur ($\sigma=2.0$)	0.8390	0.8956	0.9223
gaussian noise ($\sigma=0.1$)	0.8217	0.8552	0.8493
JPEG (quality 30)	0.8449	0.8981	0.8783
centre crop (keep 0.8)	0.9637	0.9673	0.9595
photometric ($\pm 20\%$, worst)	0.9868	0.9877	0.9904

Table 3: Per-transform retention at each transform’s worst parameter point (lowest L1 retention over its grid). Bold marks the best arm per row. The frequency branch wins where the spatial read is destroyed (resize, blur) but slightly regresses where its own read is confounded (block-aligned JPEG energy, broadband noise, and framing cuts).

calibration drift, consistent with the repair being bounded by the frozen head.

5 Limitations

The approach inherits the constraints of the frozen detector it repairs. Because GRACE operates in the trunk’s own feature space, the quality of the restoration is bounded by what that space can express: a corruption that collapses the features beyond recognition, or a head that cannot read the corrected directions, leaves no room for the adapter to recover. This is the deliberate price of not retraining the detector — there is efficiency gain and also a guarantee that every measured change is attributable to the corrective modules, but the ability to reshape the representation itself is forgone. Relatedly, the adapter is fundamentally a restorer and cannot exceed the clean-image score (retention ≤ 1); beating that ceiling requires the frequency branch, which in turn demands a supervised second stage and the added compute of a second read.

Both corrective modules are designed as exact identities at initialization and must learn to open their gates. This is what makes the reported numbers trustworthy, but it also means early training is deliberately conservative: a gate that never opens would leave the repair silently unused, and the strict capacity budget of the adapter ($\sim 0.3\text{M}$ parameters against a 21M trunk) risks under-representing the true degraded-to-clean mapping if it is more complex than the architecture can express.

The severity conditioning that calibrates the correction is itself a learned module. Severity is predicted rather than observed, so errors in the severity head propagate directly into how much correction the adapter and the enricher apply: a confidently wrong estimate can under-correct a badly degraded image or over-correct a lightly handled one. The frequency branch makes additional structural assumptions. Its two-band split presumes that the damage types of interest are separable by frequency, and its native-resolution read introduces an alignment ambiguity when the spatial and frequency branches observe different windows of the image.

6 Future Work

The most immediate extension is to the evaluation itself. A broader benchmark spanning more generators, real-world capture-and-redistribution pipelines, and a per-generator “hard” validation split would tighten the confidence intervals around the headline numbers, and reporting false-positive rates at realistic deployment base rates would characterize behaviour where AI-generated content is rare and the cost of a mistake is high.

On the method side, the scalar severity estimate could be replaced by a richer, self-supervised description of degradation — for example a learned mixture over degradation prompts — so the modules condition on the *kind* of damage as well as its amount, reducing the reliance on a single secondary model. The frequency branch invites further study of learnable band structures, top- k coefficient selection, and conditioning the native-resolution read on the crop geometry to resolve the cross-branch alignment ambiguity; a single end-to-end stage that co-adapts the adapter and the enricher may also outperform the current two-stage schedule.

Finally, the repair is detector-agnostic by construction and should be ported to other frozen backbones and

heads to test how much of the observed robustness gain transfers. Quantifying the overhead of the second read is a prerequisite for real-time deployment, and the per-channel gates of both modules offer an interpretability output that could explain, per image, which degradation types drive the detector’s decision.

7 Conclusion

We studied whether the collapse of AI-generated image detectors under real-world image handling is repairable without retraining. On a frozen DINOv3 detector we showed that a small, label-free gated residual adapter (GRACE) – identity at initialization, conditioned on a predicted severity – restores degraded features and lifts retention on the held-out COCO/DALL-E 3 benchmark from 0.75 to 0.87 at the deepest composition level, and that a frequency branch reading the same image in a per-block DCT basis pushes the repair further (clean AUC 0.972, L3 retention 0.886) precisely where the spatial read is destroyed, such as downscaling and blur. The corrected detector also behaves as if degradations compose independently, largely eliminating the interaction effect that makes single-transform benchmarks overstate the baseline’s robustness. The repair is not free: it inherits the frozen head’s blind spots, cannot exceed the clean-feature ceiling without the supervised frequency stage, and leaves a residual bias toward false positives on smoothed reals and false negatives on noised fakes. These results suggest that robustness to ordinary image handling is recoverable from a clean-trained detector by small, identity initialized corrective modules, and point to richer severity conditioning and cross-branch alignment as the natural next steps.

References

- [1] O. Siméoni et al., “DINOv3,” arXiv preprint arXiv:2508.10104, 2025.
- [2] WildFake: A large-scale dataset for AI-generated image detection, ModelScope (Apache 2.0).
- [3] A. Gushchin et al., “NTIRE 2026 Challenge on Robust AI-Generated Image Detection in the Wild,” arXiv preprint arXiv:2604.11487, 2026.
- [4] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville, “FiLM: Visual Reasoning with a General Conditioning Layer,” arXiv preprint arXiv:1709.07871, 2017.